

H1

ANDERSON JUNIOR COLLEGE
HIGHER 1
ANSWERS

2018 JC2 PRELIMINARY EXAMINATIONS

CANDIDATE
NAME

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PDG

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INDEX NUMBER

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H1 BIOLOGY

8876/02

Paper 2 Structured and Free-Response Questions

**11 SEPT 2018
TUESDAY**

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name and PD group on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graph
Do not use paper clips, highlighters, glue or correction fluid.

Answer all questions in the space provided on the Question Paper.

Section A

Answer **all** questions

Section B

Answer any **one** question

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1	/13
2	/7
3	/10
4	/11
5	/4
6/7	/15
/60	

Answer **all** questions

- 1 Fig. 1.1 shows the production of all blood cells from Cell X.

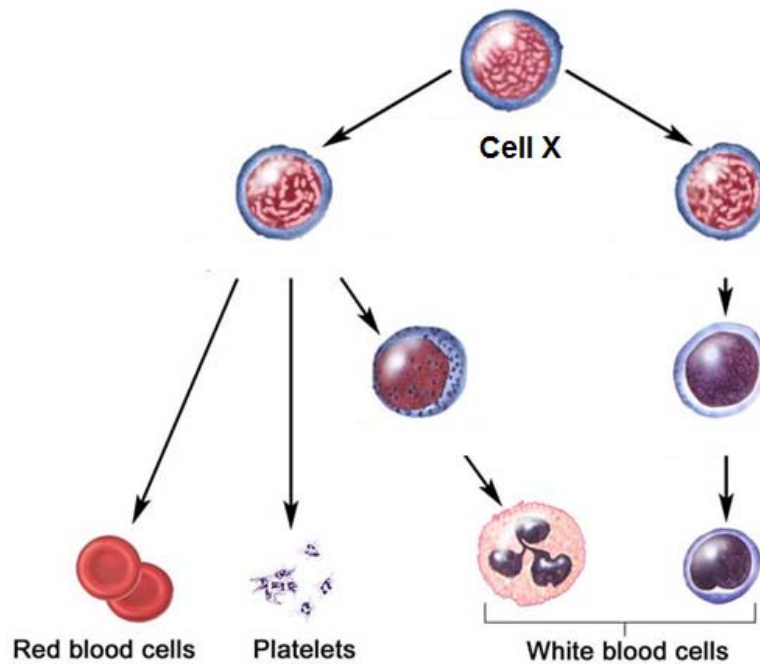


Fig. 1.1

- (a) Identify Cell X and explain why, as shown in Fig. 1.1, it must have the characteristic of a stem cell.

- Cell X: **haematopoietic stem cell / blood stem cell**;
- must be **unspecialized** / no specialised cell structure and function and **multipotent**
- so that it can **differentiate** into specialized cells such as red blood cells
- must be able to divide by **mitosis / self renew**
- to give rise to more cells which can replace worn-out / damaged blood cells;

[4]

- (b) Fig 1.2 is an electron micrograph of a type of white blood cell known as plasma cell, in the process of producing antibodies. Antibodies are proteins secreted by plasma cells.

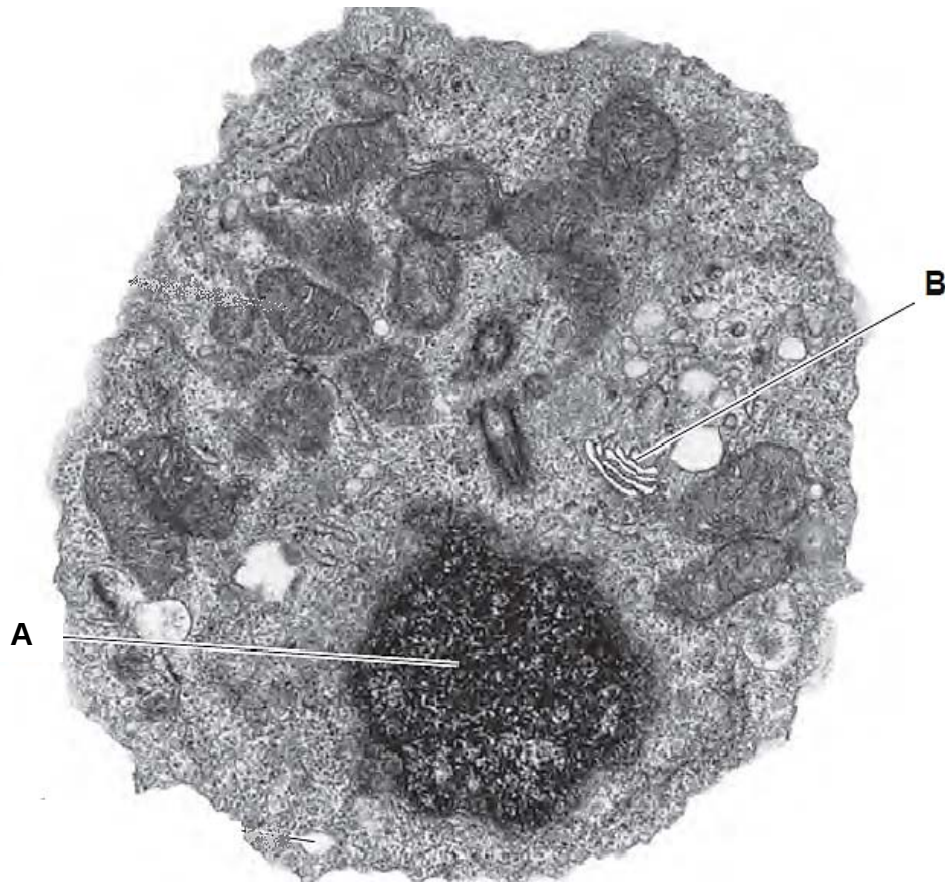


Fig. 1.1

Describe the role of the following organelles in the production of antibodies.

A:

- **Gene** encoding for antibody is found on **DNA** in the **nucleus**
- which is **transcribed** into **mRNA** in the **nucleus**

B:

- proteins are **modified, sorted** and **packaged**
- into **secretory vesicles** which **bud/ pinch off** from Golgi body.

[4]

- (c) Macrophage is another type of white blood cell which is involved in recognising foreign particles and undergoing phagocytosis.

(i) Describe the process of phagocytosis.

- (When particles bind to the cell surface membrane), **the cell surface membrane form pseudopodia** (singular form: pseudopodium) to **engulf** the particle.
- This process **requires energy** in the form of ATP.
- The tips of the pseudopodia **fuse** and **pinch off** to form phagocytic vacuole./ vesicle

[3]

(ii) Suggest how a macrophage recognises specific foreign particles.

- **receptors/ proteins** on the **cell surface membrane** of macrophages
- are **complementary in shape** to (part of) foreign particles , hence **bind** to the foreign particle

[2]

[Total: 13]

2 Fig. 2.1 shows translation occurring on a eukaryotic mRNA.

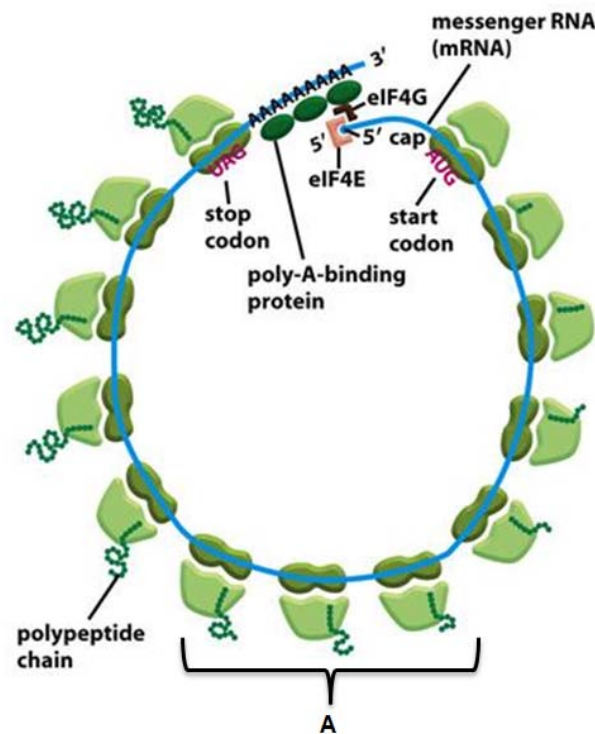


Fig. 2.1

(a) (i) Explain the significance of the pattern of translation, labelled **A** in Fig. 2.1.

- **Polyribosomes / multiple ribosomes** work on translating **one** messenger RNA
- **single** mature mRNA is used to make **many copies of the same type of polypeptide simultaneously** / enables a cell to synthesise **many copies of the same type of polypeptide very quickly**

[2]

(ii) During initiation of translation, the poly-A tail and 5'cap on the mRNA, together with initiation factors like eIF4E and eIF4G form part of a complex which aid in recruiting ribosomal subunits to mRNA.

With reference to Fig. 2.1, describe the role of poly-A tail and 5' cap in the assembly of ribosomes.

- **poly-A binding protein** binds to the **poly-A tail** which in turn recruit/ attract **eIF4G** to bind
- **eIF4E** binds to **5'cap** which **binds** to **eIF4G** (to assemble into **translation initiation complex**)

(together forming a complex which aid in recruiting ribosomal subunits to mRNA)

[3]

(b) Describe **three** differences between the process of translation and transcription.

	Transcription	Translation
Monomers	RNA nucleotides	Amino acid
Bonds formed between monomers	Phosphodiester bonds	Peptide bonds
Enzyme involved in synthesis the bond between monomers	RNA polymerase	Peptidyl transferase
Template	DNA strand	mRNA strand

[3]

[Total:7]

3 The ABO blood group system in humans is controlled by a gene with alleles, I^A , I^B and I^O . Each allele lead to production of different antigens (proteins) on the surface of red blood cells.

- Allele I^A , leads to the production of antigen A.
- Allele I^B , leads to the production of antigen B.
- Allele I^O , does not lead to the production of either antigen.

Alleles I^A and I^B are codominant and I^O is recessive to both.

(a) With reference to the ABO blood group, explain what is meant by

(i) codominance.

- Both alleles I^A and I^B are **separately** manifested/ expressed / influence blood group (distinctly)
- An individual with alleles I^A and I^B (at the gene locus controlling blood group) will have blood group AB.

[2]

(ii) multiple alleles.

- (single characteristic) Blood group is controlled by 3 or more/more than 2 alleles, I^A , I^B and I^O of the same gene/ occupying the same gene locus

[1]

(b) In humans, a gene that codes for the production of a protein, called factor VIII, is located on the X chromosome. The dominant allele for this gene produces factor VIII, but the recessive allele does not produce factor VIII.

A person who is unable to make factor VIII has haemophilia in which the blood fails to clot properly.

Explain why a man with haemophilia cannot pass haemophilia to his son but may pass haemophilia to his grandson.

- Son only inherits Y chromosome (which does not have the mutant allele) from father / do not receive X chromosome (carrying the mutant allele) from father

Some students thought the grandson mentioned in the question is the man's son's son rather than the man's daughter's son. So they talked about the man's son mating with an affected female to cause their son to have hemophilia. However this is not what the question wants since question already says "man...pass haemophilia to his grandson". The question wants you to explain the case on how the man's X-chromosome carrying the mutant allele can be passed to his grandson, rather the grandson getting the disease because of inheritance from the female that the man's son marries.

[3]

- Father will pass X chromosome carrying the mutant allele to daughters
- Daughter may pass mutant allele on X-chromosome to her son / his grandson

Reject mutated X-chromosome/ recessive X-chromosome - should be X chromosome carrying mutant allele

Reject merely saying passing down alleles without reference to X-chromosome - Must have idea of passing down X-chromosome carrying mutant allele.

Reject merely saying carrier/heterozygous without reference to X-chromosome

- (c) A gene for feather colour in chickens is carried on an autosome. This gene has two alleles, black (C^B) and splashed-white (C^W). When a male chicken with black feathers is mated with a female chicken with splashed-white feathers as shown in Fig. 3.1, all the offspring have blue feathers. This also occurs when a male chicken with splashed-white feathers is crossed with a female with black feathers.

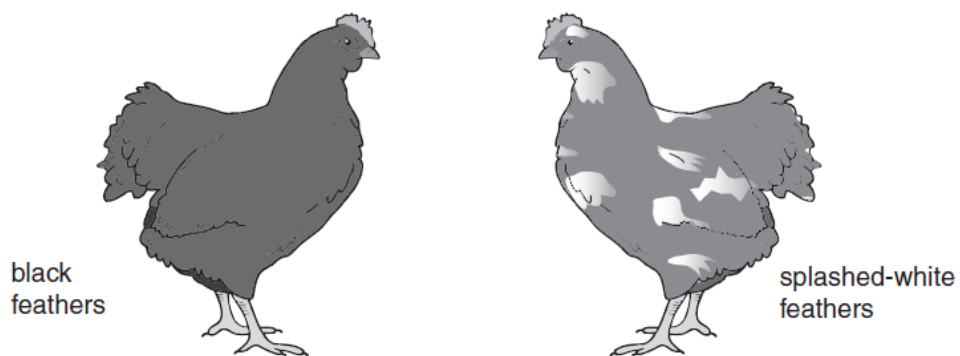


Fig. 3.1

Another gene may cause stripes on feathers (barred feathers), shown in Fig. 3.2. This gene is carried on the X chromosome. The allele for barred feathers (X^A) is dominant to the allele for nonbarred feathers (X^a).

In chickens the male is homogametic and has two X chromosomes while the female is heterogametic and has one X chromosome and one Y chromosome.

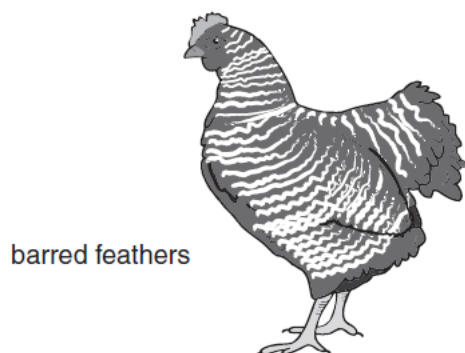


Fig. 3.2

A male chicken with black, non-barred feathers was crossed with a female chicken with splashed-white, barred feathers. All the offspring had blue feathers, but the males were barred and the females were non-barred.

Using the symbols given above, draw a genetic diagram to show this cross.

$$\begin{array}{ccc}
 \text{(male)} & C^B C^B X^a X^a & \times & \text{(female)} & C^W C^W X^A Y \\
 \text{(gametes)} & C^B X^a & & C^W X^A & \text{or} & C^W Y \\
 & C^B C^W X^A X^a & & C^B C^W X^a Y \\
 & \text{(male, blue, barred)} & & \text{(female, blue, non-barred)}
 \end{array}$$

[4]

[Total: 10]

- 4 To protect crops against insects, many crops such as maize and cotton are genetically modified to contain a gene (Bt) from a bacterium, *Bacillus thuringiensis*. The Bt gene codes for a toxin, which inserts into the insect gut cell membrane, paralyzing the digestive tract and killing the insects which eat the crop. When these genetically modified (GM) crops first became available it was predicted that insect pests would develop resistance to these toxins.

(a) Explain how Bt resistance may arise and spread in an insect population.

- Bt resistance arise as a result of: **random mutations** in the gene that encode proteins in the cell surface membrane / gene that encodes for protein that can bind to Bt to prevent it from inserting into the cell membrane / AVP in the insects

Any 2

- Presence of Bt toxin creates a **selection pressure** during **natural selection**
- Insects with resistance to Bt toxin/ not killed by Bt toxins are at a **selective advantage** / will be selected for
- can **survive, reproduce** and **pass on alleles** that confer Bt resistance to their **offspring / next generation**

[4]

- (b) The extent of Bt resistance in insect pest species was surveyed in 2005 and in 2011. The level of resistance in each species was classified according to the highest percentage of resistant individuals recorded in any population anywhere in the world. Three levels of resistance were identified:

- <1%
- 1–6%
- >50%

There were no reports of populations of insect pests having between 6% and 50% of resistant individuals.

The results of the surveys are shown in Table 4.1.

Table 4.1

year	total number of insect pest species surveyed	number of insect pest species susceptible to Bt toxins	number of insect pest species with reported levels of resistance		
			<1%	1–6%	>50%
2005	9	8	0	0	1
2011	13	4	3	1	5

- (i) Discuss whether the data provided can support the hypothesis that rise in Bt resistance is due to introduction of GM crop, including limitations of the data provided.

2 marks on data analysis

From 2005 to 2011,

- percentage of insect pest species susceptible to Bt toxins decreased from 89% to 31%

AND

- Percentage of insect pests with high level of resistance/ >50% resistance increased from 11% to 38.5%

OR

- Number of insects pests with low level of resistance / < 1% resistance increased from 0% to 23% OR Number of insects pests with low level of resistance / 1-6% resistance increased from 0% to 7%

(reject merely quoting raw values since the total number of insect pests surveyed is different- must convert to percentage)

2 marks on limitations

- Small sample size
- Survey only done in two years
- the damage done by the insect pests surveyed ;
- the number of reports of resistance for each species ;
- the proportion of populations with the highest percentage of resistant individuals ;
- the effect on the crops concerned of pest resistance at the levels given (<1%, etc.) ;e.g. the losses in yield
- the geographical spread of the insect pest species that show resistance

(ii) Justify the prediction that both insect-caused and non-insect-caused damages to Bt crops will increase with climate change.

- Extreme weather conditions- must give example e.g. drought / water logging / flood decreased growth of crops
- Increasing temperature degrade Bt toxin → Bt toxin n longer effective against insects
- Increasing temperature will increase metabolism of insects/ shorter maturation/life cycle/ reproduce in a shorten span of time → More insects to feed on crops
- increase in reproduction rate of insects , increase the development of resistance to Bt toxin

[3]

[Total: 11]

5 An analysis of ice cores from the Artic and Antarctic can provide information about the composition of the Earth's atmosphere over thousands of years.

Fig. 5.1 shows the concentrations of carbon dioxide and methane measured in ice cores, dated between 1000 and 2000 AD.

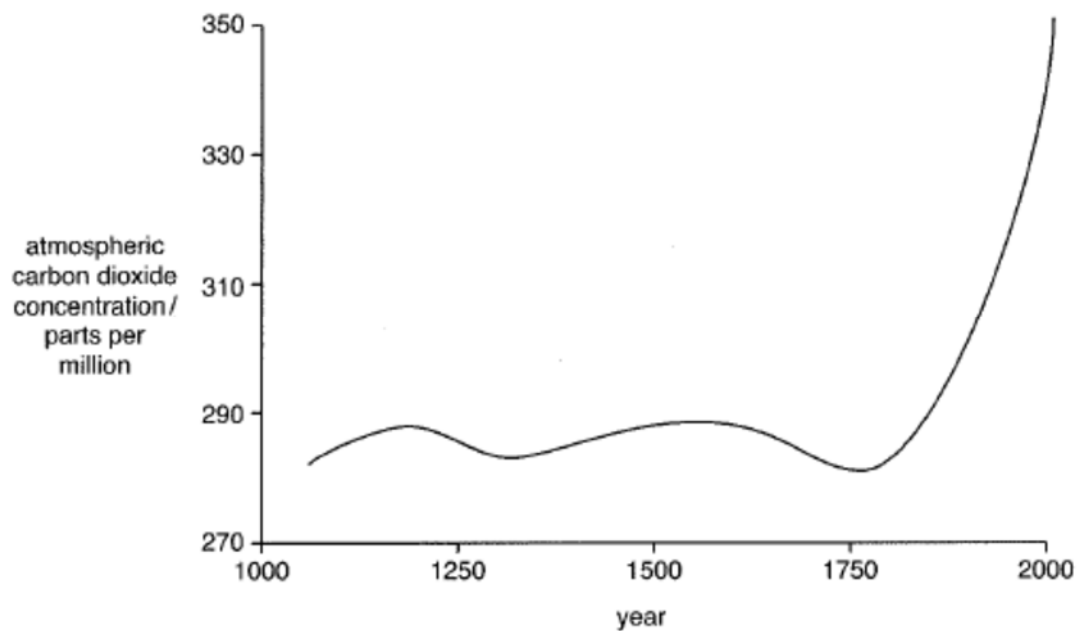


Fig.5.1

(a) Describe and explain the data in Fig. 5.1 from 1750 AD onwards. [2]

- (Describe data trend and quote) Atmospheric carbon dioxide concentration **increase sharply** from 280 to 350 parts per million from 1750 to 2000AD;
- (explain) Rise of both atmospheric carbon dioxide and methane concentration after 1750 is due to industrial revolution, rise in human population, rise in human activities, rise in demand of energy consumption, deforestation for other land uses and meat consumption;

- (b) Explain how increasing concentrations of gases, such as carbon dioxide and methane, are thought to cause global warming. [2]
- Blanket of carbon dioxide and methane around the Earth;
 - Allows high energy/ short wavelength rays from the Sun to enter Earth's atmosphere;
 - When these are reflected from the surface of the Earth;
 - They lose energy;
 - And the longer wavelength/ lower energy/ infra-red/ heat rays cannot escape through the carbon dioxide blanket;(causes global warming)

[Total: 4]

Section B

Answer **one** question in this section

Write your answers on the lined paper provided at the end of this Question Paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answer must be set out in parts **(a)**, **(b)**, etc., as indicated in the question.

- 6 (a) Explain how fluidity of biological membranes can be maintained and the importance of fluidity to membrane function. [8]

How fluidity is maintained:

- Definition of fluidity: phospholipids and proteins free to move within layer / transversely;
- Membrane fluidity can be regulated by regulating the level of cholesterol and phospholipids with unsaturated hydrocarbon tails in the membrane;
- Membranes freeze at a lower temperature if it has a higher proportion of phospholipids with unsaturated hydrocarbon tails and cholesterol; (Accept reverse argument)
- Unsaturated hydrocarbon tails have kinks that keep the phospholipid molecules in the membrane from packing close together at lower temperatures, enhancing membrane fluidity; (Accept reverse argument)
- Cholesterol prevents close packing of phospholipids in the membrane, preventing membrane from freezing at lower temperatures;
- Membrane prevented from being overly fluid / integrity of the membrane is maintained at higher temperature as cholesterol restricts phospholipid movement through its interactions with phospholipids;
- Stabilising the lipid bilayer through hydrophobic interactions;

[max 4m]

Importance:

- ref. to need for invagination / pinching in of the membrane during cytokinesis / cell division;
- ref. to invagination of the membrane during phagocytosis / pinocytosis / endocytosis;
- ref. to formation of vesicles for the transport / trafficking of proteins from rough ER to Golgi apparatus / other membrane-bound organelles;
- ref. to formation of vesicles from *trans* face of the Golgi apparatus during protein sorting;
- ref. to fusion of vesicles from endoplasmic reticulum / containing proteins to

- form cis-face of the Golgi apparatus for protein modification;
- ref. to fusion of vesicles with cell surface membrane to release secretory molecules from the cell / exocytosis;
- ref. to fusion of endocytotic vesicles with lysosomes;
- ref. membrane fluidity to allow locomotion of unicellular organisms e.g. amoeba;
- ref. diffusion of small or non-polar molecules through the membrane, e.g. water / oxygen / carbon dioxide (at least one named example);
- ref. to transmembrane transport proteins changing conformation, such as sodium-potassium pump / ligand-gated sodium ion channel / voltage-gated sodium ion channel (at least one named example);

[max 4m]

Examiners' comments:

There were many answers which incorrectly described fatty acid chains instead of fatty acid tails of phospholipids. Many students also confused phospholipids with cholesterol, describing cholesterol as have kinks or being unsaturated or saturated. There were also students who describe fatty acid chains which were branched.

Many answers included general statements about membrane functions in cells instead of relating fluidity to these functions. The general statements were not credited. Also, merely mentioning that 'membrane fluidity is important for endocytosis or exocytosis' is insufficient. It is important to explain clearly what the membrane fluidity was critical to, i.e. 'membrane fluidity is important for the invagination of the membrane during endocytosis' or 'membrane fluidity is essential for the secretory vesicles to fuse with the cell membrane during exocytosis' would be clearer answers which will gain credit.

- (b) A human body can synthesise more than 90,000 different types of proteins. The complete set of proteins present in the various cells of an individual is known as the proteome.

A human's genome is constant, whereas the proteome of a human is constantly changing.' Discuss this statement.

[7]

(discuss for human's genome is constant) due to nature of genes being DNA

1. How **semi-conservative DNA replication** before cell division takes place accurately/ described
2. **complementary base pairing** (A=T, C≡G)
3. **DNA polymerase with proofreading function**
4. Specific sequences have specific functions → important that the functions are maintained (Examples introns, exons, promoters, distal control elements (enhancer/ silencer), centromeres, telomeres)
5. one gene, one protein hypothesis
6. How mitosis ensures genetic stability/ nuclei of daughter cells are genetically identical or receive the same number and type of chromosomes as the original parent nucleus,
7. Function/ why mitosis is important: stem cells during growth of human and renewal and repair: replace dead or damaged cells by new cells which are genetically identical to original cells so that cells (used in growth/ renewal/repair) can carry out the same function;

(discuss for human's proteome is constantly changing)

8. need for changing protein expression as cells grow and specialise
9. upregulation/ down regulation of enzymes
10. temporal expression e.g. according to human developmental phase (fetal haemoglobin)/ respond to environment changes (growth factors)
11. spatial expression (different cell types requires different proteins to form different structure to perform different functions)
12. Why there is protein variety, classification of proteins/different class of protein for different functions/purpose (Examples: Transport/ enzymatic activity/ cell-to-cell recognition/ receptor binding/ cell signalling/intercellular joining/ stabilizing proteins/ respiration)
13. environmental factors (e.g. exposure to sun, increase melanin production)
14. post translational modification → alternative splicing → different mRNA (also nucleic acid) formed (support variety of proteome)

(discuss against human's genome is constant)

15. Exposure to radiation / chemical carcinogens causes structural damage to DNA/induce mutation + *cite an example below*
 - e.g. UV light (causes thymine dimer formation)
 - e.g. chemicals (such as nitrous acid chemically reacts with base)
 - e.g. ethidium bromide (intercalates into DNA)
16. **DNA replication is not always perfect** → Spontaneous mutation – DNA polymerase adds the wrong base, and is not being rectified.
17. (nature of damage) Such structural damage causes wrong nucleotide(s) / extra nucleotide(s) / missing nucleotide(s)/ change in chromosomal structure e.g. chromosomal translocation
18. Gametic cells: Generation of gametic variation as a result of meiosis (not all cells in the same human body have same genome)
19. Process that generate variation: crossing over, independent assortment of homologous chromosomes
20. Errors in mitosis leading errors in number of chromosomes e.g. aneuploidy/ mosaic Down syndrome
21. (link between genome and proteome) Proteins are derived from protein synthesis. Their unique amino acid sequences (coded for by the DNA sequence of their genes) determine their folding thus their tertiary, quaternary structure which enables them to perform their unique functions.

(discuss against human's proteome is constantly changing)

22. the need for constancy in the proteome for the proteins involved in fundamental life processes, such as respiration idea of conserved proteins/ housekeeping proteins (no matter the time/ space in a human)
23. QWC (address all sections)

Examiners' comments

Those candidates who appreciated the meaning of 'proteome' were able to engage effectively with this title and make links to a range of topic areas with which they were familiar. Stronger candidates used a clear line of explanation and argumentation to form the bulk of the essay content.

Relevant issues for consideration included the significance of mutation, the generation of variation during meiosis, various forms of genetic modification and a discussion of the possible consequences of these changes to the genome. For the proteome, candidates could have discussed the need for changing protein expression as cells grow and specialise. A small number of candidates discussed

the need for constancy in the proteome for the proteins involved in fundamental life processes, such as respiration. A discussion of conserved proteins would have improved the quality of responses of some candidates.

A few candidates misread the question and answered it in terms of 'the human genome and proteome' instead of 'a human's genome and proteome'. Although many relevant issues were still explored, it meant that the essay had an emphasis on evolutionary change in a population as opposed to the uniqueness of the individual (examiner's interpretation).

Reference to **transport**

Hydrophilic channels involved in facilitated diffusion;
For specific ions and polar molecules to pass through;
Channel proteins have fixed shape/ carrier proteins can undergo rapid changes
In shape in order to transport specific substances;
Na⁺ and K⁺ pumps involved in active transport;
Actively pumps 2K⁺ into cell and 3Na⁺ out of cell against concentration gradient;

Reference to **enzymatic activity**

Enzyme embedded in membrane with its active site exposed to substances;
E.g. inner membrane of mitochondria contains ATP synthase;
For the synthesis of ATP/ in oxidative phosphorylation;

Reference to **cell to cell recognition**

Glycoproteins serve as identification tags;
Antigens serve as markers;
To recognize specific cells for development of tissues and organs;
Reference to antibody-antigen binding
Antigens act as cell identity markers
Foreign antigens can be recognized and attacked by immune system;

Reference to **receptor binding**

Chemically signalling between cells;
Protein receptors that have binding site with specific shape;
Binds to a transmitter (hormone or neurotransmitter);
Causes conformational change to the protein;
That relays message to the inside of the cell
E.g. receptor on post-synaptic membrane/ receptors for cyclic AMP/ receptors
for signal transduction pathway/ receptors for hormonal regulation (insulin
receptors);
protein phosphorylation cascade in cell signalling

Reference to **intercellular joining**

Proteins in adjacent cells are hooked together in various kinds of junctions;
Tight junction: membrane fused to prevent leakage of fluid;
Anchoring junction: fasten cells together into strong sheets;
Gap junctions: provide channels between cells for passage of ions and small
molecules;

Reference **to stabilizing proteins**

Carbohydrate chains of glycoproteins and glycolipids project into aqueous
environment surrounding cell;
Form hydrogen bonds with water molecules which stabilize the membrane

- 7 (a) Explain why more energy is released from the oxidation of triglycerides as compared to the same unit mass of glucose and outline the problems of using triglycerides as the main respiratory substrate instead of glucose in animals. [7]

- triglycerides contain **two times more hydrogen atoms / greater ratio of hydrogen to oxygen atoms / more C-H bonds** as compared to the same mass of glucose;
- Can be used to generate **greater** number of **NADH** and **FADH** during **Kreb's cycle / dehydrogenation reactions**;
- Release **more** high energy **electrons** to the electron transport chain;
- Generate more ATP by **oxidative phosphorylation**;
- Glucose is **polar / hydrophilic** due to presence of many hydroxyl groups;
- Glucose is therefore soluble in aqueous medium, can easily be transported in blood and cellular cytoplasm;
- Small molecule so can diffuse through blood vessels / across the cell surface membrane through the glucose carrier proteins / via facilitated diffusion;
- Triglycerides are **non-polar / hydrophobic** due to the fatty chains / hydrocarbon chains;
- Triglycerides are insoluble in aqueous medium, cannot be easily transported in aqueous medium / needs to form lipoprotein complexes with proteins to be transported in blood;
- Cannot easily cross the cell surface membrane / needs to enter the cell via **receptor-mediated endocytosis**;
- AVP;

QWC (one point from each section);

- (b) Discuss the inter-relationships of membranous organelles in eukaryotic cells and suggest the implications of absence of membranous organelles in prokaryotic cells. [8]

Protein synthesis and secretion

- **Nucleus** : site of **transcription** (of DNA to form RNA)
- Using **RNA nucleotides/ RNA polymerase** and **DNA template** found in the nucleus
- RNA (accept merely mention mRNA?) formed leave via **nuclear pores** to enter **cytoplasm**

Ignore reference to free ribosomes

- **Ribosomes** found on **rough endoplasmic reticulum** carry out **translation** of proteins of the endomembrane system OR proteins to be secreted from the cell
- Route from protein: Proteins are then packaged into **transport vesicles** → **Golgi body**
- Within the Golgi body, proteins from RER are further **modified, sorted** and **packaged**
- **Lipids** from **smooth ER** are also **modified, sorted** and **packaged**
- From Golgi body to **cell surface membrane** via **secretory vesicles** / to other parts of the cell via the **Golgi vesicles**

- Award once only for **pinching off/ budding** of vesicles from organelles + **fusion** of membrane of vesicles with the destination organelle

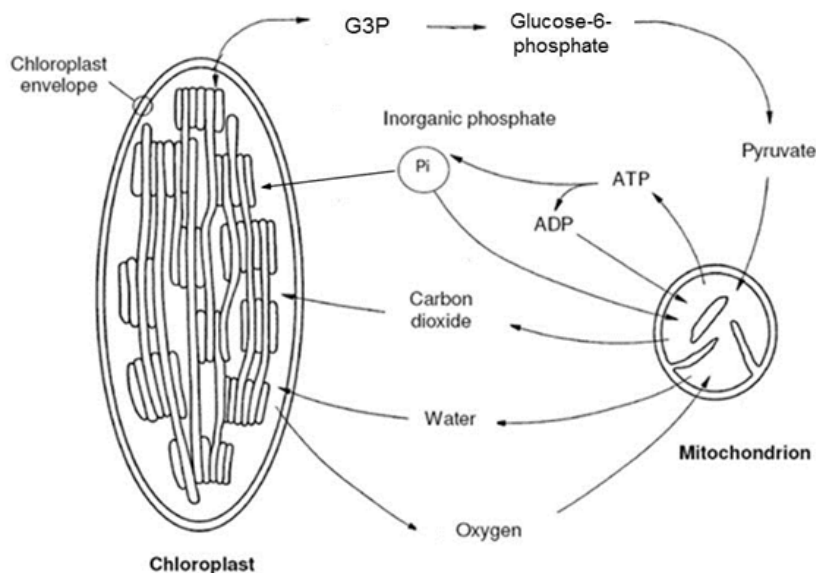
E.g. Membrane of transport vesicles fuse with the membrane (of the *cis* face) of the Golgi body and deposit their proteins within the lumen of the Golgi body OR secretory vesicles which bud/ pinch off from (the *trans* face of the) Golgi body to fuse with CSM

- award once only for movement of vesicles : With the aid of **microtubules**, the secretory vesicles will move to the cell surface membrane

Mitochondria / chloroplast with RER (TYS)

- mitochondrion **produces ATP** for rough endoplasmic reticulum for protein synthesis/translation
- RER synthesises proteins/ enzymes which are required for respiratory reactions in mitochondria/ photosynthesis in chloroplast

Mitochondria and chloroplast (2013 Promos) in plant cells



- **G3P** formed in **chloroplast** due to **light-independent reaction/ Calvin cycle** (converted to glucose)
- **pyruvate** (accept glucose?) **undergo aerobic respiration** / the **link reaction, Krebs cycle and oxidative phosphorylation** to be broken down for **ATP production in mitochondria**
- Oxygen is formed (In thylakoid space) as a **product of the photolysis of water** during **light-dependent reaction**
- Oxygen (diffuses out of the chloroplast and into the mitochondrial matrix) acts as the **final electron acceptor** (in the electron transport chain to produce water) in oxidative phosphorylation
- Carbon dioxide **produced** in aerobic respiration during **Krebs cycle and link reaction** / by oxidative decarboxylation
- (Diffuses into the stroma of the chloroplast) for **carbon fixation** in the **light independent reaction/Calvin cycle**;

Lysosomes

- Vesicles containing **digestive/ hydrolytic enzymes**
- Which **bud off / pinch off** from the **Golgi apparatus**
- **Phagocytosis**: Substances taken in by the cell in a vacuole/ **phagocytic vesicles** are fused with lysosomes
- **Autophagy**: A damaged organelles or small amount of cytosol becomes surrounded by a membrane, which then fuses with lysosome to form a vacuole in which the unwanted materials are digested.

Implication of absence of membranous organelles in prokaryotes

- DNA **not enclosed** by **nuclear membrane**
- This allows **transcription** and **translation** to take place **simultaneously**
- No RNA processing (since transcription and translation occurs simultaneously)
→ no **post-transcriptional control** of gene expression
- **Fewer genes / smaller genome**: since there is no need to encode transport proteins/ signaling proteins/ etc on membranes of organelles
- Protein secretion (no exocytosis/ endocytosis) can only occur through **channel proteins / facilitated diffusion**
- Stalked particles / respiration / photosynthesis (for cyanobacteria) occurs only on **cell surface membrane**